Vyshnavi DP

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Summary

Software engineer with **6+ years of production experience** who builds Python-based **LLM applications, RAG workflows, and developer tools**. Shipped Dell SupportAssist's AI assistant and built **Sentinel**, an AI code-review GitHub App with hybrid retrieval, schema-validated output, and a 98-fixture evaluation harness. Combines agent evaluation and generated-code review with secure backend services, CI testing, production troubleshooting, and cross-functional delivery in C#/.NET and Java.

Professional Experience

Dell Technologies via Accenture **Aug 2021 – Jul 2024**

Senior Software Engineer *Bangalore, India*

- Initiated, designed, and built the **in-app AI assistant** in Python and React, and shipped it to end users. It answers questions using **RAG** and the PC's diagnostics, warranty, and service history, and can start scans, open cases, or hand off to a human.
- Cut status requests by about **60%** by replacing client polling with **WebSocket** updates from the C# service for diagnostic scans that run 15 to 20 minutes, and made progress resumable so a client that disconnects mid-scan picks up where it left off.
- Cut the regression cycle from **3 days to 8 hours** by introducing **Cypress** end-to-end testing and owning it across the application, and wrote the first CI/CD deployment scripts for the new cloud version on Docker and Kubernetes (AWS).
- Designed and built features in the **C#/.NET Windows service** behind SupportAssist, Dell's diagnostics agent preinstalled on **10M+ PCs**: a local HTTPS API restricted to Dell origins, scheduled scans, and signed repair scripts.
- Built the **telemetry pipeline** that sends disk health, temperature, voltage, and crash data from **2M+ active devices** to the cloud, with batching, encryption, and on-device anonymization, feeding predictive battery and disk failure alerts.
- Worked on the **AngularJS to Angular migration** of the SupportAssist client, **mentored junior engineers**, and served in the **on-call rotation** for production services, triaging incidents and driving fixes.

Tata Consultancy Services **Jun 2018 – Aug 2021**

Software Engineer *Chennai, India*

- Kept product search under **200 ms P95** across **2M+ SKU records** for retailers including Woolworths and Action, using Java and Spring Boot services with Redis caching and parallel queries.
- Built core services for Macrospace, Optumera's space planning module, which turns **1,000+ business parameters** into recommendations for **1,400+ stores**, along with D3.js dashboards that show optimization progress live over SSE.
- Proposed **multi-tenant SSO** after noticing customers had to sign in separately to each module they bought, then built it with Spring Security, OAuth2, JWT, and Redis so one login covers all of a customer's modules.

Research Experience

San José State University (Agentic AI Systems) **Sep 2025 – May 2026**

Research Assistant *San Jose, CA*

- Built a Python **agent evaluation harness** comparing LLaMA-3 and Mistral on tool-call accuracy, JSON validity, and error recovery. Used step traces and failure categories to guide prompt and tool-interface improvements.
- Investigated a **semantic-saturation stopping criterion** for question generation from document corpora, testing whether smaller question sets preserve coverage for **RAG retrieval evaluation** with Prof. Bill Andreopoulos.

Projects

Sentinel | LLM code review | *Python, FastAPI, PostgreSQL, pgvector* **GitHub**

- Implemented GitHub App integration with HMAC-verified webhooks, delivery deduplication, schema-validated model output, and hybrid full-text/pgvector retrieval with reciprocal rank fusion.
- Built a **98-fixture evaluation harness** using synthetic code diffs and authored expected review comments. Scored per-category precision, recall, and F1, and categorized failures to diagnose missed issues and false positives.
- Developed Sentinel with **Claude Code and Cursor**: wrote task specifications, reviewed generated diffs and agent transcripts, and checked results against expected review comments.

Technical Skills

AI Agents | LangChain, LangGraph, planner-executor workflows, tool calling, schema validation, retries
LLMs and RAG | RAG, hybrid retrieval, pgvector, LLM evaluation, benchmark design, precision/recall/F1
Developer Workflows | Git/GitHub, code review, Claude Code, Cursor, generated-diff review, agent trace analysis
Languages | Python, Java, C#, TypeScript, Go, JavaScript, SQL
Backend and Delivery | FastAPI, Spring Boot, .NET, PostgreSQL, Redis, AWS, Docker, Kubernetes, GitHub Actions
Frontend and Testing | React, Angular, Cypress, Jest, JUnit

Education

San José State University **Aug 2024 – May 2026**

M.S. in Software Engineering *San Jose, CA*

- **AgentHacks 2025:** LLM Preference Aware System using Python, Streamlit, and NetworkX.
- **Teaching Assistant:** Deep Learning and Data Mining; authored grading rubrics. Coursework: Distributed Systems, Machine Learning, Deep Learning, NLP.